Drafts of the GitHub Project

Vladyslav Demydiuk

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1 General

2 Introduction

Take a look at the following quantum circuit, with the somewhat simplified Dicke states for 4 qubits and their subsequent measurement.

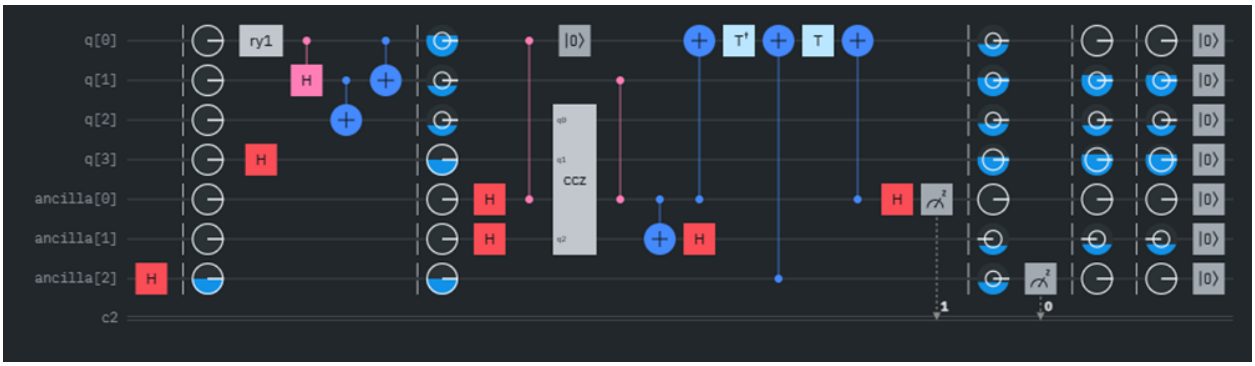

Figure 1: Basic Oracle Construction

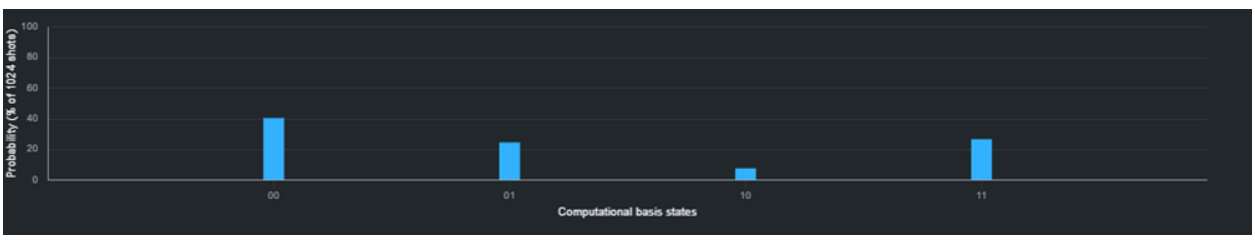

Figure 2: The Corresponding Simulated Probabilities

The probabilities from left to right are $41\frac{2}{3}\%$, 25%, $8\frac{1}{3}\%$, 25%. If the qubit ancilla[0] is 0, then the probability of ancilla[2] being 1 is $\frac{25\%}{41\frac{2}{3}\%+25\%} * 100\% = 37.5\%$. And if the qubit ancilla[0] is 1, then the probability of ancilla[2] being 1 is 75%. (We will return to these probabilities in Section 4.6 later).

The qubit ancilla[0] stores some information on whether it was measured on q[0] or not, which was, in turn, dependent on the ancilla[2]. In other words, based on the measurement result of ancilla[0], one can determine the state of ancilla[2] more or less accurately.

There is no causal connection between ancilla[0] and ancilla[2], though — they do not exchange information through the X-gates.

This article shows that the probabilities in the histogram above can be derived via a hypergeometric model, using Bayes' theorem.

3 Background

3.1 Rules

The article presents calculations for the hypergeometric distribution with a deck of cards, black (successes) and red.

We analyze the case where the following is done with the N-card deck ($\frac{N}{2}$ of them black):

1. Half of the deck is drawn; black cards are counted
2. The agent decides whether half of the rest of the cards should be drawn or not
3. Half of the rest of the cards are either drawn or not; black cards are counted

We assume that the setting of the experiment satisfies certain conditions that we will go through in the section 4. If so, then after step 1 is done, one can derive the final posterior probability that the experiment will continue based only on the number of black cards in the first draw, using the Bayesian theorem. In fact, the greater the deviation of the first sample is, the bigger the probability that the experiment was continued.

3.2 Definitions

4 Hypergeometric Sampling

4.1 Formalities

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

With the prior probability of continuation being set to 0.5, we get:

$$P(Continued|k_0 = x) = \frac{P(k_0 = x|Continued) * 0,5}{P(k_0 = x)}$$

$$P(k_0 = x) = h(x, n, M, N) = \frac{(\frac{M}{x})(\frac{N-M}{n-x})}{\frac{N}{n}} = \frac{(\frac{0.5N}{x})(\frac{0.5N}{0.5N-x})}{\frac{N}{0.5N}} = \frac{(\frac{0.5N}{x})(\frac{0.5N}{0.5N-x})}{2}$$

$$P(k_0 = x|Continued) = \sum_{i=0}^{\frac{1}{4}N} \sum_{j=0}^{\frac{1}{2}N} a_{ij}$$

$$a_{ij} = P(both\ trials\ like\ this) * P(second\ trial\ like\ this)$$

$$P(both\ trials\ like\ this) = \min\left(P(X \leq \mathbb{E}[X]), P(X \geq \mathbb{E}[X])\right), where\ X = k_1 i + k_0 j$$

$$\&\& \ where \ \mathbb{E}[X] = \frac{\frac{3}{4} * \frac{N}{2}}{N}$$

Of course, the cumulative distribution function is discrete and can be noted as follows:

$$P(X \geq \mathbb{E}[X]) = 1 - (F_{\mathbb{E}[X]}((k_1 i + k_0 j - 1); (N - (N * 0.5 * 0.5)); N/2; N))$$

$$= 1 - (\sum_{y=0}^{k_{1i}+k_{0j}-1} \frac{(\frac{0.5N}{y-1})(\frac{N-0.5N}{\frac{3}{4}N-(y-1)})}{\frac{N}{\frac{3}{4}N}}) = 1 - (\sum_{y=0}^{k_{1i}+k_{0j}-1} \frac{3}{4}(\frac{0.5N}{y-1})(\frac{0.5N}{\frac{3}{4}N-y+1}))$$

$$P(X \leq \mathbb{E}[X]) = (F_{\mathbb{E}[X]}((k_{1i} + k_{0j}); (N - (N * 0.5 * 0.5)); N/2; N))$$

$$= \sum_{y=0}^{k_{1i}+k_{0j}} \frac{(\frac{0.5N}{y})(\frac{N-0.5N}{\frac{3}{4}N-y})}{\frac{N}{\frac{3}{4}N}} = \sum_{y=0}^{k_{1i}+k_{0j}} \frac{3}{4}(\frac{0.5N}{y})(\frac{0.5N}{\frac{3}{4}N-y})$$

$$P(second \ trial \ like \ this) = h(k_{1i}, \frac{N-0,5N}{2}, \frac{N}{2} - x, N - \frac{N}{2})$$

$$= \frac{(\frac{0.5N-x}{k_{1i}})(\frac{0.5N-(0.5N-x)}{0.25N-k_{1i}})}{\frac{0.5N}{0.25N}} = \frac{(\frac{0.5N-x}{k_{1i}})(\frac{x}{0.25N-k_{1i}})}{2} = (\frac{0.25N-x}{k_{1i}})(\frac{x}{0.25N-k_{1i}})$$

4.2 Vandermonde's identity logic

Now let's take a short detour:

$$\sum_{k=0}^{r} \binom{m}{k}\binom{n}{r-k} = \binom{m+n}{r}$$

4.3 Simplifying the final formula[1]

$$P(Continued|k_0 = x) =$$

$$\frac{\sum_{k_{1i}=0}^{\frac{1}{4}N} \sum_{k_{0j}=0}^{\frac{1}{2}N} (min[(1-(\sum_{y=0}^{k_{1i}+k_{0j}-1} \frac{3}{4}(\frac{0.5N}{y-1})(\frac{0.5N}{\frac{3}{4}N-y+1})));(\sum_{y=0}^{k_{1i}+k_{0j}} \frac{3}{4}(\frac{0.5N}{y})(\frac{0.5N}{\frac{3}{4}N-y}))]*(\frac{0.25N-x}{k_{1i}})(\frac{x}{0.25N-k_{1i}}))*0.5}{\frac{(\frac{0.5N}{x})(\frac{0.5N}{0.5N-x})}{2}}$$

$$= \frac{\sum_{k_{1i}=0}^{\frac{1}{4}N} \sum_{k_{0j}=0}^{\frac{1}{2}N} (min[(1-(\sum_{y=0}^{k_{1i}+k_{0j}-1} \frac{3}{4}(\frac{0.5N}{y-1})(\frac{0.5N}{\frac{3}{4}N-y+1})));(\sum_{y=0}^{k_{1i}+k_{0j}} \frac{3}{4}(\frac{0.5N}{y})(\frac{0.5N}{\frac{3}{4}N-y}))]*(\frac{0.25N-x}{k_{1i}})(\frac{x}{0.25N-k_{1i}}))}{(\frac{0.5N}{x})(\frac{0.5N}{0.5N-x})}$$

$$= \frac{\sum_{k_{1i}=0}^{\frac{1}{4}N} \sum_{k_{0j}=0}^{\frac{1}{2}N} (min[(1-(\sum_{y=0}^{k_{1i}+k_{0j}-1} \frac{3}{4}(\frac{0.5N}{y-1})(\frac{0.5N}{\frac{3}{4}N-y+1})));(\sum_{y=0}^{k_{1i}+k_{0j}} \frac{3}{4}(\frac{0.5N}{y})(\frac{0.5N}{\frac{3}{4}N-y}))]*(\frac{0.25N-x+x}{0.25N}))}{(\frac{0.5N}{x})(\frac{0.5N}{0.5N-x})}$$

[1]Nothing in this article relies on these simplifications

4.4 Key claim

Let 'z' denote the Confidence Interval (CI) of the first sample, where the standard deviation δ_0 is a single unit: $z = \sqrt{\frac{(x-0.25*N)(\sqrt{N-1})}{N}}$. Then, $P(Continued|k_0 = x)$ is approximately a function of 'z' (due to the Central Limit Theorem).

4.5 Visualisation via MS Excel

The greater N is, the smaller the deviations:

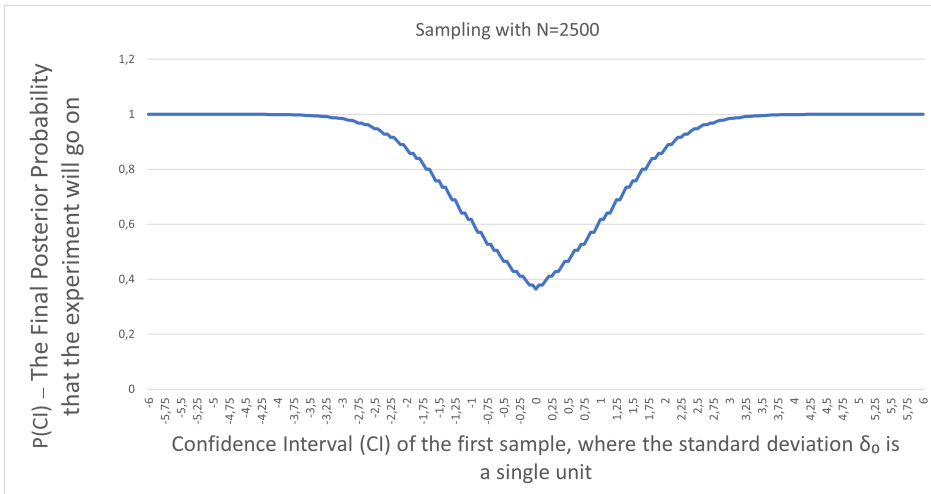

Figure 3: Sampling graph with N = 2500

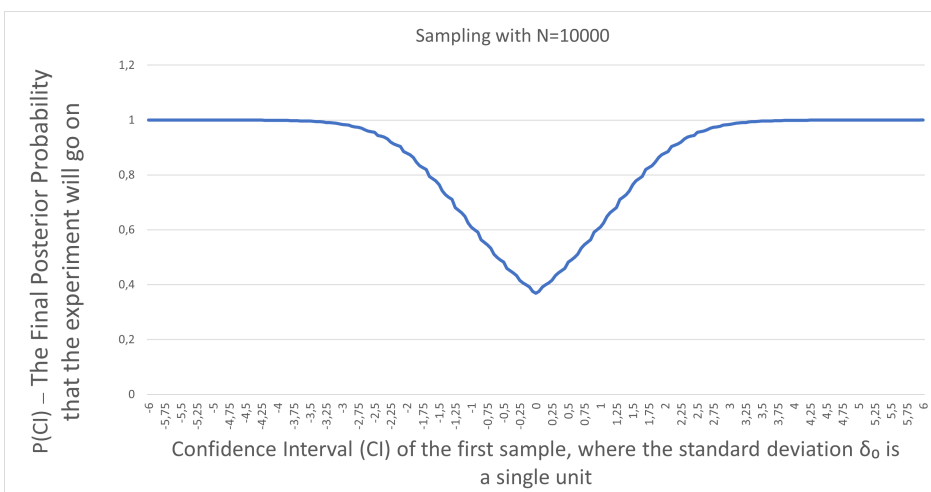

Figure 4: Sampling graph with N = 10 000

From the graph, we can see that the distribution can be described through a function $f(x) : y = \frac{|x|^{\frac{3}{2}}}{a\ -\ b*|x|^{\frac{3}{2}}} + 1 - \frac{1}{b}$, where b might be close to 1.6 and a to 2.5.

4.6 Sampling with N = 4

This is the minimum population that we need for our experiment. The probabilities for $N = 4$ are the same as in the histogram above (see Introduction): 0.75 for 0, 0.375 for 1, and 0.75 for 2.

4.7 The Key Takeaway

If we expect to get some fixed amount of information from an experiment, and the results are enriched, then the experiment was continued rather than not. If the results are de-enriched, then we would guess that the experiment was stopped.